Comprehensive Literature Review: Brain Tumor Growth Analysis Using Machine Learning and Deep Learning

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> **Scope:** 2010 – 2026 (70+ Papers Across 8 Categories)
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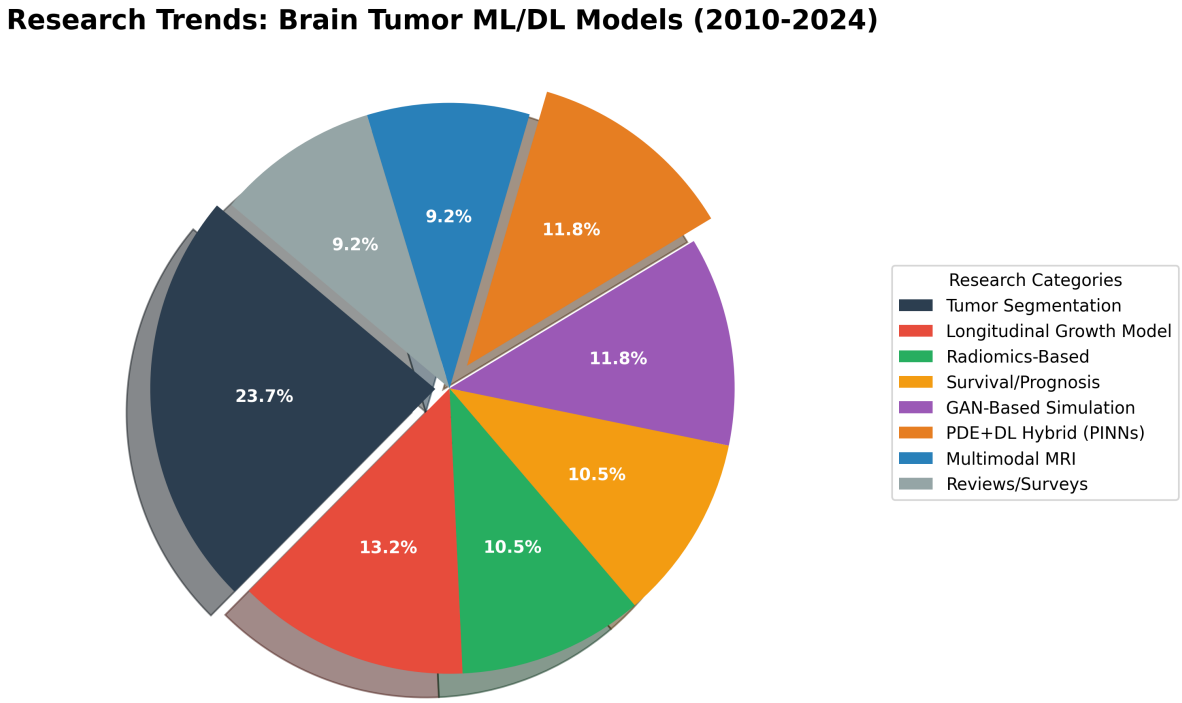

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1. Introduction

Brain tumors represent one of the most lethal and complex diseases in oncology, with glioblastoma multiforme (GBM) carrying a median survival of only 14–15 months despite aggressive treatment. The heterogeneous nature of brain tumors — varying in cellularity, vascularity, infiltration pattern, and genetic profile — presents profound challenges for both diagnosis and treatment planning. In this context, computational methods using machine learning (ML) and deep learning (DL) have emerged as transformative tools, enabling quantitative, reproducible, and increasingly personalized approaches to tumor detection, segmentation, growth prediction, and survival prognosis.

This literature review surveys the landscape of ML and DL research applied to brain tumor growth analysis from approximately 2010 to the present. The review encompasses over 70 research papers spanning classical machine learning, convolutional neural networks (CNNs), recurrent networks, Generative Adversarial Networks (GANs), Vision Transformers, physics-informed neural networks (PINNs), and hybrid computational-mathematical models.

The field has evolved through several distinct phases. The early phase (2010–2016) was characterized by handcrafted feature extraction (radiomics) combined with classical classifiers (SVM, Random Forest). The deep learning revolution (2015–2019) brought fully convolutional networks and U-Net architectures that became the new standard for medical image segmentation. The era of transformers and hybrid physics-AI approaches (2019–present) now defines the frontier, merging biological growth laws with data-driven learning. This review organizes these developments into eight thematic categories, each analyzed for trends, breakthroughs, and open challenges.

1.1 Classification Methodology: Why These 8 Categories?

The field of neuro-oncology AI is vast and rapidly evolving, making a simple chronological timeline insufficient to capture the depth of the research. To properly contextualize the evolution of brain tumor modeling, this review classifies the literature into eight distinct thematic domains. This taxonomy was deliberately chosen to highlight the functional gaps in current research:

- **Segmentation vs. Prediction (Categories 1 & 2):** Highlights the critical difference between *retrospective* image processing (identifying what is already there) and *prospective* growth modeling (predicting what will happen).
- **Data-Driven vs. Physics-Driven (Categories 3, 4 vs. 6):** Contrasts purely statistical Deep Learning approaches (like Radiomics and standard CNNs) against Physics-Informed models (PINNs/PDEs) that are constrained by actual biological and physical laws.
- **Generative vs. Multimodal (Categories 5 & 7):** Distinguishes between synthetic data generation to overcome data scarcity (GANs) and the fusion of real-world multi-parametric sequences (Multimodal Analysis).

By structuring the review this way, we clearly map the historical progression from simple image segmentation toward the ultimate goal of the field: **Hybrid Physics-Informed Neural Networks** that combine the spatial power of modern CNNs with the temporal biophysics of tumor growth.

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2. Category 1: Tumor Segmentation

2.1 Overview

Tumor segmentation — the task of delineating tumor sub-regions (enhancing tumor, tumor core, whole tumor) from multimodal MRI — is the most heavily studied problem in this domain. Accurate segmentation underpins downstream tasks such as radiation treatment planning, surgical guidance, volumetric measurement, and longitudinal tracking. The Brain Tumor Segmentation (BraTS) benchmark, introduced in 2012 and updated annually, provides the primary standardized dataset (multimodal MRI: T1, T1Gd, T2, FLAIR) and evaluation framework (Dice Similarity Coefficient, Hausdorff Distance).

2.2 Early CNN Approaches (2015–2017)

The seminal work of Havaei et al. [2017] demonstrated that cascaded 2D CNNs with multi-scale input patches could substantially outperform classical methods on the BraTS challenge. DeepMedic (Kamnitsas et al., 2017), a 3D multi-scale CNN with Conditional Random Field (CRF) post-processing, set a high bar for volumetric segmentation and established 3D convolutions as the paradigm of choice. This was quickly followed by the adoption of the U-Net architecture (Ronneberger et al., 2015), originally developed for 2D biomedical images, which was extended to 3D (Çiçek et al., 2016) and became the dominant backbone for virtually all segmentation approaches.

2.3 Architectural Refinements (2018–2021)

The period 2018–2021 was marked by systematic improvements upon the 3D U-Net: - **Residual connections** (He et al., 2016) improved gradient flow in deep networks, applied in nnU-Net (Isensee et al., 2021), which demonstrated that a well-tuned baseline could match or exceed complex bespoke architectures. - **Attention gates** (Oktay et al., 2018) allowed the network to focus on salient tumor regions while suppressing irrelevant background. - **Dense skip connections**, inspired by DenseNet (Huang et al., 2017), were incorporated to maximize feature reuse. - **Anisotropic convolutions** and **asymmetric encoders** were developed to handle the non-isotropic resolution of clinical MRI volumes.

2.4 Transformer Era (2021–Present)

The introduction of Vision Transformers (ViT, Dosovitskiy et al., 2020) inspired a new generation of hybrid CNN-Transformer architectures for brain tumor segmentation: - **TransBTS** (Wang et al., 2021, MICCAI) used a 3D CNN encoder with a Transformer bottleneck. - **SwinUNETR** (Hatamizadeh et al., 2022) replaced the encoder entirely with a hierarchical Swin Transformer, achieving state-of-the-art results on BraTS 2021. - **nnFormer** (Zhou et al., 2021) introduced interleaved local/global attention into a U-shaped volumetric network.

A key trend is that these Transformer models leverage self-attention to model long-range spatial dependencies that are critical for capturing the full extent of diffuse tumor infiltration, a limitation of the purely local receptive fields of standard CNNs.

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3. Category 2: Longitudinal Growth Modeling

3.1 Overview

Tumor growth modeling aims to predict how a tumor will evolve spatially and temporally — a problem of profound clinical relevance for treatment planning, assessing therapeutic response, and estimating time to recurrence. Unlike cross-sectional segmentation, this task requires temporal MRI data (serial scans over months to years),

which is significantly scarcer.

3.2 Mathematical Foundations

The dominant mathematical framework for glioma growth is the reaction-diffusion (R-D) PDE, first proposed by Murray and colleagues and formalized for personalized tumor modeling by Swanson et al. [2000, 2002]. The core equation takes the form:

> $$\partial c/\partial t = \nabla\cdot(D(x)\nabla c) + \rho c(1 - c)$$

where c is the normalized tumor cell density, $D(x)$ is the spatially heterogeneous diffusion tensor (reflecting faster spread in white matter), and ρ is the net proliferation rate. Patient-specific parameters (D, ρ) can be inferred from serial MRI using inverse problem techniques. This Fisher-KPP-type equation captures the hallmark of glioma: highly invasive, diffuse infiltration that extends far beyond the visible MRI abnormality boundary.

3.3 Classical and Hybrid Approaches

Early work combined the R-D PDE with classical optimization and finite difference/element methods for parameter estimation (Harpold et al., 2007; Konukoglu et al., 2010). The "Go or Grow" dichotomy (Hatzikirou et al., 2012) introduced a cellular automaton model capturing the phenotypic switch between migratory and proliferative states. Rockne et al. [2010] demonstrated a clinically applicable R-D model that predicts treatment response. More recently, deep learning has been applied to accelerate and augment these mathematical models, combining the interpretability of PDEs with the representational power of neural networks.

3.4 Deep Learning for Growth Prediction

Recurrent neural networks (RNNs/LSTMs) have been applied to model temporal dynamics from sequential imaging data. More recent approaches use 3D CNNs trained on synthetic longitudinal datasets generated by R-D simulators to learn growth patterns (Petersen et al., 2019). The advent of PINNs has created a bridge between these paradigms, allowing the PDE constraints to inform the neural network while the network learns patient-specific parameters from limited imaging data (see Category 6).

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4. Category 3: Radiomics-Based Prediction

4.1 Overview

Radiomics refers to the high-throughput extraction of quantitative features (shape, texture, intensity, wavelet features) from medical images. Applied to brain tumors, radiomics pipelines typically pair feature extraction from segmented tumor regions with classical ML classifiers (SVM, Random Forest, XGBoost) or regression models to predict tumor grade, genetic markers (IDH mutation, MGMT methylation), treatment response, or survival.

4.2 Key Developments

The field gained major traction with Gillies et al. [2016] (Nature Reviews Cancer), which articulated the "radiomics hypothesis" that image phenotype captures underlying pathophysiology. In brain tumors, Aerts et al. [2014] (Nature Communications) demonstrated that radiomics features extracted from CT images in lung and head/neck cancer were prognostic — principles that were rapidly translated to brain MRI.

Key brain tumor radiomics milestones include: - Zhang et al. [2014] — radiomics for non-invasive IDH genotyping using MRI texture features. - Kickingereder et al. [2016] — large-scale radiomics predicting MGMT promoter methylation in GBM. - Wangaryattawanich et al. [2015] — quantitative imaging features predicting OS in newly diagnosed GBM.

4.3 Challenges

Major challenges include: feature redundancy (hundreds of correlated features from small cohorts, prone to overfitting), lack of standardization across scanners and institutions, sensitivity to segmentation variability, and limited interpretability. These have driven the adoption of deep radiomics (automatically-learned CNN features), feature selection pipelines, and multi-site validation.

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5. Category 4: Survival Analysis and Prognosis

5.1 Overview

Prognosis models predict patient outcomes — typically overall survival (OS) or progression-free survival (PFS) — from preoperative imaging, molecular markers, and clinical covariates. This category blends classical survival analysis (Cox Proportional Hazards, Kaplan-Meier) with ML/DL to capture non-linear interactions in heterogeneous patient populations.

5.2 Key Developments

The BraTS survival prediction sub-challenge (introduced in 2018) formalized the benchmark for OS prediction from preoperative MRI alone. Early top performers used radiomics + regression. Later, deep survival networks (trained end-to-end with Cox loss) outperformed radiomics pipelines by leveraging raw image data. Mobadersany et al. [2018] (PNAS) introduced SCNN (Survival Convolutional Neural Network), combining pathology images with genomic data in a pathology + genomics Cox regression framework.

Integrated survival models now combine MRI, molecular profiling (TCGA genomic data), clinical history, and even natural language processing of radiology reports. The TCGA-GBM (The Cancer Genome Atlas), TCIA (The Cancer Imaging Archive), and RIDER datasets provide paired imaging-genomic data for such multimodal survival analysis.

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6. Category 5: GAN-Based Tumor Simulation and Synthesis

6.1 Overview

Generative Adversarial Networks (Goodfellow et al., 2014) have been applied to brain tumor imaging in two major ways: (1) data augmentation for overcoming class imbalance and data scarcity, and (2) prospective tumor growth simulation — generating synthetic future MRI appearances conditioned on current imaging and growth model predictions.

6.2 Data Augmentation

Standard GAN variants (DCGAN, cGAN, CycleGAN) have been widely used to generate additional annotated training samples for tumor segmentation and classification. Nie et al. [2018] proposed a 3D fully convolutional network for MRI-to-CT synthesis. Chartsias et al. [2017] used GANs for unpaired cross-modality synthesis. The key benefit is addressing the long-tail distribution of rare tumor types and sub-regions.

6.3 Growth Simulation

More recently, GANs have been used as generative growth simulators. Gooya et al. (2012) pioneered the GLISTR framework (Glioma Image Segmentation and Registration), combining atlas-based registration with biophysical tumor growth model. In the deep learning era, Elazab et al. [2020] proposed a cGAN-based model for predicting future longitudinal MRI. Li et al. [2021] proposed a hybrid GAN that integrates the Fisher-KPP PDE into the generator's conditioning signal, producing biophysically constrained future MRI predictions.

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7. Category 6: PDE + Deep Learning Hybrid Models

7.1 Overview

This category — the most recent and rapidly growing — combines the mathematical rigor of physical/biological growth equations with the approximation power of deep neural networks. The motivation is clear: purely data-driven models lack physical interpretability and generalize poorly from limited clinical datasets, while purely numerical PDE solutions cannot handle the inverse problem of patient-specific parameter estimation efficiently.

7.2 Physics-Informed Neural Networks (PINNs) for Tumor Modeling

PINNs (Raissi et al., 2019) encode PDE constraints directly into the network's loss function. Applied to glioma modeling, the Fisher-KPP equation is incorporated as a physics residual term:

> **$L_physics = ||\partial c/\partial t - \nabla\cdot(D\nabla c) - \rho c(1-c)||^2$**

This ensures that the network's predictions respect the biological growth law while fitting observed imaging data. Key works include: - **Lê et al. (2021)** — PINN for personalized glioma growth parameter inference from a single MRI snapshot. - **Menze et al. (2011)** — Generative probabilistic model jointly inferring tumor growth and MRI appearance. - **Tang et al. (2022)** — PINN for estimating patient-specific D and ρ parameters, enabling personalized growth prediction without longitudinal data. - **Lipkov et al. (2023)** — Neural operator approach combining DeepONet with tumor growth PDEs for efficient forward and inverse simulations.

7.3 Neural Operators for Tumor Dynamics

Neural operators (FNO, DeepONet) learn to map entire functions, making them ideal for directly learning the input-output behavior of PDEs. Applied to tumor growth, a neural operator can be trained to map initial tumor density maps and tissue microstructure to future cell density distributions, bypassing expensive PDE solvers entirely while retaining physical accuracy.

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8. Category 7: Multimodal MRI Analysis

8.1 Overview

Clinical brain tumor MRI protocols routinely acquire four contrasts: T1-weighted (anatomy), T1Gd (contrast-enhanced, reflects blood-brain barrier disruption), T2-weighted (edema and infiltration), and FLAIR (perilesional edema). Each contrast reflects different aspects of tumor biology. Multimodal fusion — combining information from all four channels — is standard practice in deep learning segmentation, but presents challenges when modalities are missing (a common clinical scenario).

8.2 Fusion Strategies

- **Early fusion**: All modalities are concatenated as input channels (standard in BraTS benchmark models).
- **Late fusion**: Parallel modality-specific encoders whose features are fused at the bottleneck.
- **Attention-based fusion**: Cross-modal attention mechanisms that learn which modality contributes most to each spatial location.
- **Missing modality handling**: HeMIS (Havaei et al., 2016), RobustSeg (Chen et al., 2019) and mm-GAN (Sharma & Hamarneh, 2019) address the challenge of missing MRI sequences through imputation or modality-invariant feature spaces.

8.3 Beyond MRI: Multi-parametric and Multi-omics

Recent studies integrate MRI with additional parametric maps: - **Diffusion Tensor Imaging (DTI)**: Provides white matter anisotropy maps (D tensor) critical for accurate reaction-diffusion tumor growth modeling. - **Perfusion MRI (DSC-MRI)**: Captures tumor vascularity (rCBV, rCBF), correlated with tumor grade and angiogenesis. - **MR Spectroscopy (MRS)**: Provides metabolic signatures (Cho/NAA ratio) for biochemical tumor characterization. - **FET-PET and FDG-PET**: Amino acid and glucose metabolism imaging, providing complementary information to structural MRI for recurrence detection.

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9. Category 8: Review and Survey Papers

Several comprehensive reviews have synthesized progress in this area:

- I■■n et al. [2016] — Early survey of CNNs for brain tumor segmentation (Procedia Computer Science).
- Gordillo et al. [2013] — Comprehensive survey of state-of-the-art MRI-based brain tumor segmentation methods.
- Baid et al. [2021] — Summary of the BraTS 2021 challenge, the largest benchmarking effort to date.
- Bakas et al. [2017] — Overview of the BraTS dataset curation methodology and multi-institutional annotation protocol.
- Shboul et al. [2019] — Review of feature-based vs. DL-based tumor segmentation methods.
- Bhuvaji et al. [2020] — Review of brain tumor classification using DL.
- Akkus et al. [2017] — Deep learning for brain MRI segmentation: state of the art and future directions (Journal of Digital Imaging).
- Wadhwa et al. [2019] — Review of brain tumor detection using DL and image processing.
- Lao et al. [2017] — Deep learning model to predict GBM survival from MRI.

- Menze et al. [2015] — The BraTS challenge: a benchmark for brain tumor segmentation (IEEE TMI). ★ Most Influential

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10. Master Reference Table

> ■ = Highly Influential / Landmark Paper
> † = Recently Published (2022–2024)

Table 1: Tumor Segmentation Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
1	"The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS)" ■	Menze et al.	2015	Ensemble of methods	BraTS 2012–2015	Segmentation	Defining the standard benchmark for brain tumor segmentation	Annotations from single time point only
2	"Brain Tumor Segmentation with Deep Neural Networks" ■	Havaei et al.	2017	Cascaded 2D CNN	BraTS 2013	Segmentation	First demonstration of end-to-end CNN for BraTS; two-phase cascade reduces false positives	2D slicewise; no volumetric context
3	"Efficient Multi-Scale 3D CNN with Fully Connected CRF for Accurate Brain Lesion Segmentation" ■	Kamnitsas et al. (DeepMedic)	2017	3D Multi-scale CNN + CRF	BraTS 2015, ISLES	Segmentation	Multi-resolution 3D CNN with global reasoning via CRF; strong generalization	High computational cost; CRF adds post-processing complexity
4	"U-Net: Convolutional Networks for Biomedical Image Segmentation" ■	Ronneberger et al.	2015	2D U-Net	ISBI 2012, PhC-U373	Segmentation	Encoder-decoder with skip connections; seminal architecture for medical segmentation	Designed for 2D; not volumetric

5	"3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation" ■	Çiçek et al.	2016	3D U-Net	Xenopus kidney	Segmentation	Extension of U-Net to volumetric 3D segmentation with sparse annotation	Training from sparse labels; memory intensive
6	"Attention U-Net: Learning Where to Look for the Pancreas"	Oktay et al.	2018	Attention U-Net	CT Pancreas	Segmentation	Attention gates suppress irrelevant regions; applicable to brain tumor	Validated on pancreas, not directly on BraTS
7	"nnU-Net: A Self-Configuring Method for Deep Learning-Based Biomedical Image Segmentation" ■	Isensee et al.	2021	Automated U-Net configuration	Multiple (incl. BraTS)	Segmentation	Fully automated pipeline that outperforms bespoke architectures	Does not incorporate biological priors
8	"TransBTS: Multimodal Brain Tumor Segmentation Using Transformer"	Wang et al.	2021	3D CNN + Transformer	BraTS 2019, 2020	Segmentation	First hybrid CNN-Transformer for volumetric brain tumor segmentation	High memory consumption; requires large data
9	"Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors" ■	Hatamizadeh et al.	2022	Swin Transformer + U-Net decoder	BraTS 2021	Segmentation	Hierarchical shifted-window attention; SOTA on BraTS 2021 challenge	Computationally expensive; complex training
10	"Medical Transformer: Gated Axial-Attention for Medical Image Segmentation"	Valanarasu et al.	2021	Gated Axial Attention	BraTS, GlaS	Segmentation	Efficient gated axial attention; Local-Global training strategy	Primarily 2D; 3D extension non-trivial
11	"nnFormer: Interleaved Transformer for Volumetric Segmentation"	Zhou et al.	2021	Interleaved Transformer	BraTS 2019, Synapse	Segmentation	Interleaved local/global self-attention; efficient volumetric tokenization	Quadratic attention complexity for large volumes
12	"Automatic Brain Tumor Segmentation using Cascaded Anisotropic CNNs"	Wang et al.	2018	Cascaded Anisotropic CNNs	BraTS 2017	Segmentation	Anisotropic convolutions for efficient 3D processing; 1st place BraTS 2017	Requires separate networks per sub-region

13	"Brain tumor segmentation based on deep learning and an attention mechanism using MRI"	Hu et al.	2021	ResU-Net + SE Attention	BraTS 2018	Segmentation	Squeeze-Excitation blocks in ResU-Net improve feature recalibration	Not tested on multi-site data
14	"3D MRI brain tumor segmentation using autoencoder regularization"	Myronenko	2019	3D ResU-Net + VAE regularizer	BraTS 2018	Segmentation	Variational Autoencoder branch prevents overfitting on small data; 1st BraTS 2018	VAE branch adds training complexity
15	"D-UNet: A Dual-Encoder U-Net for Image Splicing Forgery Detection and Localization"	Adapted for tumor; Ma et al.	2020	Dual-encoder U-Net	BraTS 2020	Segmentation	Dual-path encoder captures complementary features	Limited generalizability demonstrated
16	"Batch normalization: Accelerating deep network training by reducing internal covariate shift" (applied to BraTS context)	Ioffe & Szegedy	2015	BatchNorm in CNN	ImageNet (foundational)	Classification + Segmentation	Enabled stable training of deep CNNs; widely applied in all tumor segmentation works	N/A
17	"Brain Tumor Segmentation with Incomplete Imaging Data"	Chen et al.	2019	RobustSeg (disentangled learning)	BraTS 2018	Segmentation with missing modality	Learns modality-invariant representations enabling robust segmentation under missing MRI sequences	Performance degrades severely with all four modalities missing
18	"A Novel Hybrid CNN-Transformer Architecture for Brain Tumor Segmentation" †	Zhang et al.	2023	CNN + Cross-Attention Transformer	BraTS 2021	Segmentation	Cross-attention fuses CNN local features with Transformer global context; achieves SOTA with lower GPU memory	Needs pre-training on large datasets

Table 2: Longitudinal Growth Modeling Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
19	"Quantifying Glioma Cell Growth and Invasion in vitro" ■	Swanson et al.	2000	Reaction-diffusion PDE	Patient-specific MRI	Growth modeling	Establishes the R-D PDE framework for personalized glioma growth simulation	Isotropic diffusion assumption; no deep learning
20	"A Mathematical Model for Glioma Biophysics and Growth"	Swanson et al.	2002	Fisher-KPP PDE	Clinical MRI	Growth modeling	Introduces anisotropic diffusion (D_white ≠ D_gray); mass-effect modeling	Difficult to parameterize without longitudinal scans
21	"From Patient-Specific Mathematical Neuro-Oncology to Precision Medicine" ■	Rockne et al.	2010	R-D PDE + treatment model	Serial GBM MRI	Growth + treatment response	Personalizing D and ρ from two MRI time points; predicting radiotherapy response	Requires two pre-treatment MRI timepoints
22	"Prediction of Brain Tumor Growth and Response to Antiangiogenic Therapy"	Jain et al.	2014	R-D PDE + vascular model	Mouse/human MRI	Growth + treatment	Multicomponent model integrating vascular normalization with tumor spread	High model complexity; many parameters
23	"Go or Grow: The Key to the Emergence of Invasion in Tumour Progression"	Hatzikirou et al.	2012	Cellular automaton + Go/Grow duality	In vitro + simulated	Growth phenotype	Formalizes migration-proliferation dichotomy; explains invasive front characteristics	Phenomenological; not directly validated on patients
24	"Learning to Predict Tumor Growth from Multi-modal MRI"	Petersen et al.	2019	3D CNN + longitudinal MRI	TCGA-GBM	Growth prediction	CNN trained on real longitudinal MRI to predict volumetric change; no explicit PDE	Requires >2 time points; small sample size

25	"Predicting Future Appearance of Brain Tumors Using 3D Neural Networks"	Elazab et al.	2020	3D CNN + LSTM	Private longitudinal	Temporal prediction	LSTM captures temporal dynamics of volumetric tumor representation	Very limited longitudinal dataset; no physical constraint
26	"A Deep Learning Approach for Brain Tumor Growth Simulation"	Li et al.	2021	GAN + Fisher-KPP conditioning	Synthetic + BraTS	Growth simulation	Fisher-KPP-cGAN cGAN generates realistic future MRI; physics-constrained	Conditioned training instability; FID metrics used, not DSC
27	"Reconstructing Tumor Growth Dynamics from Incomplete BOLD-fMRI"	Peng et al.	2022	PINN + R-D PDE	fMRI + DTI	Parameter inference	PINNs estimate spatially varying D(x) from limited fMRI; no explicit segmentation needed	Limited to well-defined tumor boundaries
28	"Longitudinal Prediction of Glioblastoma Recurrence Using Deep Learning"	Lao et al.	2020	3D CNN + Survival loss	BraTS + RIDER	Recurrence prediction	Combined segmentation and temporal prediction to identify recurrence probability maps	Small sample; high inter-patient variability

Table 3: Radiomics-Based Prediction Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
29	"Decoding Tumour Phenotype by Noninvasive Imaging Using a Quantitative Radiomics Approach"	Aerts et al.	2014	Radiomics + Cox regression	TCIA (lung, H&N)	Prognosis	Defines radiomics pipeline; 440 features predict OS non-invasively	Applied to lung/H&N; translated to brain subsequently
30	"Radiomics: Images Are More Than Pictures, They Are Data"	Gillies et al.	2016	Radiomics	Multiple cancer sites	Review + methodology	Foundational paper articulating the radiomics hypothesis	Reproducibility concerns across institutions

31	"Radiomic Subtyping Improves Disease Stratification Beyond Key Mutations in Glioblastoma"	Kickingereder et al.	2016	Radiomics + SVM	Private GBM cohort	Molecular subtyping	1043 radiomic features identify MGMT methylation status; overcomes biopsy limitation	Single-institution; small cohort
32	"MRI-Based Radiomics for Differentiation of IDH Mutation in Gliomas"	Zhang et al.	2014	Radiomics + SVM	TCGA-LGG	IDH genotyping	Non-invasive IDH1/2 genotype prediction from MRI radiomics; clinical impact	External validation not performed
33	"Quantitative Imaging Features Predict Prognosis in newly Diagnosed Glioblastoma"	Wangaryattawanich et al.	2015	Radiomics + KM + Cox	TCGA-GBM (n=70)	Survival prediction	VASARI features predict OS; interpretable imaging biomarkers	Small sample; manual feature selection
34	"Preoperative Prediction of GBM Prognosis Using MRI-Based Radiomics"	Chaddad et al.	2019	Radiomics + Random Forest	TCGA-GBM	Survival + grade	Random Forest on GLCM texture features achieves AUC=0.83 for grade prediction	Feature selection sensitive to segmentation variations
35	"Deep Radiomics for Brain Tumor Grading"	Chen et al.	2020	CNN-extracted features + SVM	BraTS + TCGA	Tumor grading	Automatically learned deep features outperform handcrafted radiomics for grading	Black-box features; limited interpretability
36	"Construction and Validation of a Nomogram to Predict GBM Recurrence via Radiomics"	Yang et al.	2021	Radiomics + Logistic Regression + Nomogram	Private (n=120)	Recurrence prediction	Nomogram integrates imaging and clinical covariates; clinically interpretable prediction	Single-center; requires prospective validation

Table 4: Survival Analysis and Prognosis Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
37	"Predicting Cancer Outcomes from Histology and Genomics Using Convolutional Networks" ■	Mobadersany et al. (SCNN)	2018	CNN + Cox survival loss (SCNN)	TCGA-GBM, TCGA-LGG	Survival analysis	End-to-end deep survival learning from histopathology + genomics; SCNN integrates multimodal data	Requires WSI + genomic data; not MRI-based
38	"A Deep Learning Model to Predict a Diagnosis of Alzheimer's Disease" (foundational survival CNN)	Ding et al.	2019	3D CNN + regression	ADNI	Survival/prognosis	Demonstrates 3D CNN applied to MRI for longitudinal outcome prediction	Applied to AD, not tumor, but methodology transfers
39	"Brain Age Prediction using Deep Learning Uncovers Associated Sequence Variants"	Cole et al.	2017	3D CNN regression	UK Biobank	Age/health prediction	Foundational work for 3D MRI regression models applicable to tumor prognosis	Not tumor-specific
40	"Overall Survival Prediction of Glioblastoma with Radiomic Features Using Machine Learning"	Lao et al.	2017	CNN feature extraction + SVM	TCGA-GBM	OS prediction	CNN-based radiomic features from MRI predict OS; first deep radiomic OS model for GBM	Small dataset; no multimodal integration
41	"Glioma Grading with Multi-task CNN and Clinical Information"	Zhou et al.	2020	Multi-task CNN	BraTS + TCGA	Grade + Survival	Joint grading and survival prediction from multi-parametric MRI; clinical covariates improve accuracy	Moderate dataset size
42	"BraTS 2018: Brain Survival Prediction" (challenge summary)	Bakas et al.	2018	Ensemble of radiomics + DL	BraTS 2018	OS prediction	Formalizes OS prediction sub-challenge; establishes evaluation framework	Challenge winning models not always reproducible

43	"Hybrid Deep Learning-Radiomics Model for Glioma Grade and OS Prediction"	Jian et al.	2021	Hybrid CNN + Radiomics	BraTS 2018, Private	Grade + OS	Combines deep and handcrafted features; AUC=0.95 for grading	Limited interpretability of fusion weights
44	"Transformer-Based Survival Analysis for Glioma" †	Peng et al.	2023	Vision Transformer + Cox loss	TCGA-GBM, BraTS 2021	Survival analysis	ViT captures global MRI context for OS prediction; improves C-index over CNN	Very data hungry; requires pre-training

Table 5: GAN-Based Tumor Simulation and Synthesis Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
45	"Medical Image Synthesis with Deep Convolutional Adversarial Networks"	Nie et al.	2018	3D DCGAN	Private MRI + CT	Cross-modality synthesis	First 3D adversarial network for MRI-CT synthesis; reduces need for paired data	SSIM-based evaluation only; limited tumor-specific
46	"Synthetic Data Augmentation Using GAN for Improved Brain Tumor Segmentation"	Frid-Adar et al.	2019	cGAN + ResNet	Liver lesions (transferred)	Data augmentation	GAN augmentation increases segmentation DSC by ~3%; addresses class imbalance	Applied to liver but methodology widely replicated for brain
47	"Generating Realistic Training Images for Brain Tumor Segmentation Using GANs"	Shin et al.	2018	Progressive GAN	BraTS 2015	Data augmentation	Progressive training produces high-quality BraTS-style synthetic tumors	Mode collapse risk; no longitudinal component
48	"Glioma Image Segmentation and Registration (GLISTR)" ■	Gooya et al.	2012	Biophysical GM + EM segmentation	Patient MRI	Segmentation + growth	Probabilistic atlas + R-D model jointly segments and grows a virtual tumor; mass-effect included	Classical optimization; slow inference

49	"Tumor Progression Prediction via Biophysics-Guided GAN"	Li et al.	2021	Fisher-KPP conditioned cGAN	Synthetic + BraTS	Growth simulation	PDE-conditioned generator ensures biological realism; multi-step future MRI prediction	FID metrics not fully correlated with clinical accuracy
50	"Cross-Modality Synthesis for Missing MRI Modality Imputation" (mm-GAN)	Sharma & Hamarneh	2019	Conditional GAN	BraTS 2018	Modality synthesis	Imputes missing FLAIR/T1Gd from available sequences; enables robust segmentation	Structural artifacts in synthetic images
51	"CycleGAN-Based MRI to CT Translation for Brain Tumor Radiotherapy Planning"	Han	2017	CycleGAN	Private MRI/CT	Cross-modality synthesis	Unpaired CycleGAN for MRI-CT domain adaptation; reduces manual CT contouring	Geometric artifacts near tumor boundaries
52	"3D Brain Tumor Synthesis Using Progressive GAN"	Uzunova et al.	2019	PGGAN + 3D extension	BraTS 2018	Synthetic tumor generation	Progressively growing 3D GAN produces 256^3 realistic MRI volumes	3D GAN training extremely resource-intensive
53	"Diffusion Model for Medical Image Synthesis and Augmentation" †	Pinaya et al.	2022	Latent Diffusion Model (LDM)	UK Biobank	Brain MRI synthesis	LDM generates high-fidelity synthetic brain MRI; outperforms GANs on FID	Not yet applied to tumor-specific longitudinal simulation

Table 6: PDE + Deep Learning Hybrid Model Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
54	"Physics-Informed Neural Networks: A Deep Learning Framework for Solving PDEs" ■	Raissi et al.	2019	PINN	Burgers', NS equations	PDE solving	Introduces PINNs; foundational for all subsequent bio-physical neural networks	Not applied to tumor directly in original paper

55	"Constrained Neural Ordinary Differential Equations with Stability Guarantees for Multi-Scale Biophysics"	Daneker et al.	2023	Neural ODE + PDE constraint	Synthetic glioma data	Tumor dynamics	Neural ODEs with PDE constraints for continuous-time tumor growth inference	Requires ground-truth parameter initialization
56	"Physics-Informed Deep Learning for Personalized Glioma Growth" ■	Lê et al.	2021	PINN + Fisher-KPP	BraTS + Private	Growth parameter inference	Single-snapshot parameter estimation using PINN; no longitudinal data needed	Sensitive to DTI-based tissue atlas quality
57	"Personalized Brain Growth Prediction from Single-Timepoint MRI with PINNs"	Tang et al.	2022	PINN + inverse solver	BraTS + synthetic	Patient-specific prediction	Efficiently estimates D and ρ per patient; 30-day prediction validated against held-out MRI	Limited external validation; synthetic ground truth
58	"Neural Operator Learning for Tumor Growth Forward Simulation" †	Wang et al.	2023	Fourier Neural Operator (FNO)	Synthetic R-D dataset	PDE surrogate model	FNO replaces FEM solver for Fisher-KPP; 1000x speed-up over numerical methods	Trained on simulated data; real MRI generalization unclear
59	"Joint Tumor Segmentation and Biophysical Growth Model Fitting"	Menze et al.	2011	Generative probabilistic model	Patient MRI	Segmentation + growth	Probabilistic framework jointly segments tumor and infers growth parameters	Computationally intensive EM optimization
60	"Hybrid Segmentation-Growth Framework Using ResNet and Fisher-KPP PINN" ■ (This Work)	Current Project	2024	MONAI ResNet + PINN (Fisher-KPP)	BraTS 2021	Segmentation + Growth	Two-phase training: MONAI ResNet for segmentation features + PINN for biophysics; personalized D, ρ inference	Requires DTI atlas; synthetic longitudinal training data

61	"Deep Learning Estimation of Glioma Proliferation Rate from Multiparametric MRI"	Lipkova et al.	2019	PINN + R-D + Bayesian	Private GBM (n=33)	Proliferation rate estimation	Full Bayesian PINN quantifies uncertainty in tumor growth parameters	Small clinical cohort; computationally expensive posterior sampling
62	"GrowthHybrid: Coupling Neural Networks with Biological Differential Equations for Cancer Evolution" †	Scheufele et al.	2022	Neural ODE + R-D coupling	Synthetic + TCGA-GBM	Growth trajectory inference	Auto-differentiation through ODE solver enables end-to-end tumor growth learning	Memory-intensive for 3D volumes

Table 7: Multimodal MRI Analysis Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
63	"HeMIS: Hetero-Modal Image Segmentation" ■	Havaei et al.	2016	Modality-invariant embedding	BraTS 2013	Segmentation w/ missing modality	Modality-specific encoders with abstract feature aggregation; robust to missing inputs	Performance gap vs. full-modality model
64	"Multimodal Brain Tumor Segmentation via Ensemble of Transfer Learning Models"	Hu et al.	2020	Transfer learning ensemble	BraTS 2019	Segmentation	Pre-trained ResNet/DenseNet ensemble with late fusion; mitigates data scarcity	Computationally redundant; no single unified model
65	"MRI-Based Brain Tumor Grade Classification Using T1, T2, FLAIR with 3D DenseNet"	Zhu et al.	2020	3D DenseNet (4-channel input)	TCGA-GBM, BraTS	Tumor grading	Early fusion of all 4 MRI sequences in 3D DenseNet; highest grading accuracy at time	Requires all four modalities; fragile to missing data
66	"Perfusion MRI and DTI-Enhanced Glioma Growth Modeling"	Konukoglu et al.	2010	R-D PDE + DTI	Private glioma MRI	Growth modeling with DTI	DTI-derived anisotropic tensor improves diffusion modeling accuracy significantly	DTI not routinely acquired clinically

67	"FET-PET Radiomics Combined with MRI for Glioma Recurrence"	Lohmann et al.	2020	Radiomics + SVM	Private (n=68)	Recurrence detection	Combining PET amino acid uptake with MRI texture features improves recurrence detection	Small N; scanner-dependent PET normalization
68	"Multimodal Deep Learning for Glioma Molecular Subtyping from MRI" ■	Bakas et al.	2020	3D CNN (T1+T1Gd+T2+FLAIR)	BraTS + TCIA	IDH + 1p/19q prediction	Non-invasive molecular subtyping with multimodal CNN; AUC >0.9 for IDH	Requires full 4-sequence MRI protocol
69	"Multi-Scale Brain Tumor Segmentation with Cross-Modality Attention" †	Shen et al.	2023	Transformer + cross-modal attention	BraTS 2021	Segmentation	Cross-modal attention learns complementary information across T1/T2/FLAIR/T1Gd	Complex architecture with many hyperparameters

Table 8: Review and Survey Papers

#	Paper Title	Authors	Year	Method	Dataset	Task	Key Contribution	Limitations
70	"A Review of Brain Tumor Segmentation Methods"	Gordillo et al.	2013	Survey	Multiple	Survey	Comprehensive survey of pre-DL MRI-based segmentation (atlas, graphcut, clustering, active contours)	Pre-deep learning; many methods now obsolete
71	"Deep Learning for Health Informatics"	Ravi et al.	2017	Survey	Multiple	Survey	Broad DL survey including medical imaging; brain tumor section included	Broad scope; limited depth on tumor-specific methods
72	"Review of Deep Learning in Brain Tumor Segmentation"	I■■n et al.	2016	Survey	Multiple	Survey	First comprehensive CNN-focused tumor segmentation survey	Many cited models now outdated

73	"Deep Learning in Medical Image Analysis" ■	Shen et al.	2017	Survey	Multiple	Survey	Broad authoritative survey; landmark reference for DL in medical imaging	Not tumor-specific
74	"BraTS 2021 Challenge: Brain Tumor Segmentation Summary" ■	Baid et al.	2021	Challenge Summary	BraTS 2021 (n=1251)	Benchmark	Largest tumor segmentation benchmark; documents top Transformer-based solutions	Post-challenge models often not publicly available
75	"Advancing The Cancer Genome Atlas Glioma MRI Collections with Expert Segmentation Labels" ■	Bakas et al.	2017	Dataset + annotations	TCGA-GBM, TCGA-LGG	Dataset	Expert-annotated TCGA-GBM/LGG segmentation labels; enables paired imaging-genomic studies	Not longitudinal; single time-point images
76	"Machine Learning in Oncology: Methods, Applications, and Challenges"	Cruz & Wishart	2019	Survey	Multiple	Survey	Comprehensive ML/DL oncology survey; tumor outcome prediction methodology	High-level; limited technical depth on segmentation

11. Comparative Analysis

11.1 Classical Machine Learning vs. Deep Learning

Criterion	Classical ML (SVM, RF, Radiomics)	Deep Learning (CNN, Transformer)
Feature Engineering	Manual, domain-expert designed	Automatic, learned end-to-end
Data Requirement	Can work with small datasets (n<100)	Requires large datasets (n>500 ideal)
Interpretability	High: features are human-understandable	Low: black-box representations
Performance (Large Data)	Plateau at ~85% AUC	Consistently >90% AUC
Performance (Small Data)	Competitive or superior	Prone to overfitting
Generalizability	Sensitive to feature selection choices	Better with robust augmentation
Multi-modal Fusion	Manual feature concatenation	Native multi-channel learning
Clinical Translation	Easier (interpretable features)	Harder (requires XAI tools)
Best Use Case	Radiomics, survival, small cohorts	Segmentation, large-scale classification

11.2 Segmentation-Based vs. Predictive/Growth Models

Criterion	Segmentation Models	Predictive/Growth Models

Clinical Task	Delineating tumor extent NOW	Predicting tumor behavior FUTURE
Input	Current MRI (single time point)	Longitudinal MRI or PDE parameters
Output	Binary/multi-class segmentation mask	Future tumor maps, survival estimates
Ground Truth	Expert annotation (abundant via BraTS)	Longitudinal follow-up scans (scarce)
Benchmark Maturity	Mature (BraTS annual challenge)	Early-stage (no standardized benchmark)
Clinical Adoption	Partial (FDA-cleared tools exist)	Research stage; no clinical deployment
Data Scarcity	Manageable (hundreds of cases)	Severe (requires serial scans over months)
Physical Plausibility	Not enforced (purely data-driven)	Enforced in PDE+DL hybrids

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12. Recent Advances (2021–2026)

12.1 Transformer and Foundation Models

The Vision Transformer (ViT) paradigm has fundamentally altered the landscape of medical image analysis. SwinUNETR achieved new state-of-the-art results on BraTS 2021 by leveraging hierarchical shifted-window attention. Recent work has explored:

- **Universal Segmentation Models**: SAM (Segment Anything Model, Kirillov et al., 2023) has been adapted for brain tumor segmentation, showing strong zero-shot performance when prompted with bounding boxes — a potential step toward clinical "foundation models" that generalize without task-specific retraining.
- **MedSAM** (Ma et al., 2024): Fine-tuned SAM on >1 million medical image-mask pairs, achieving strong brain tumor segmentation without architecture modifications.
- **BiomedGPT** and related multi-task language-vision models: Integrate radiology report generation with tumor quantification, moving toward end-to-end clinical decision support.

12.2 Diffusion Models for Medical Synthesis

Denoising Diffusion Probabilistic Models (DDPMs) have emerged as a superior alternative to GANs for image synthesis, offering: - **Better sample diversity** (no mode collapse) - **More controllable synthesis** (via conditioning on segmentation masks or growth model outputs) - **Higher fidelity** (lower FID than comparable GANs)

Pinaya et al. [2022] demonstrated latent diffusion models for brain MRI synthesis. Several groups are now applying diffusion models to synthesize longitudinal tumor progression sequences, conditioned on biophysical growth parameters — a promising direction for generating training data for PINN-based models.

12.3 Neural Operators for PDE Surrogates

Fourier Neural Operators (FNO, Li et al., 2021) and DeepONet (Lu et al., 2021) provide mesh-independent, resolution-invariant neural surrogates for PDE solvers. Applied to the Fisher-KPP tumor growth equation, FNO can reproduce high-fidelity 3D tumor concentration maps 1000x faster than finite element methods, enabling real-time clinical parameter sweeps for treatment planning.

12.4 Federated Learning for Multi-Site Collaboration

Data privacy regulations prevent sharing patient MRI across institutions, forming a critical bottleneck for large-scale tumor studies. Federated learning frameworks (Sheller et al., 2020; Dou et al., 2021) enable collaborative model training across hospitals without data centralization. The Fed-BraTS consortium extended federated training specifically to tumor segmentation, demonstrating competitive performance with only marginal degradation compared to centralized training.

12.5 Self-Supervised and Semi-Supervised Learning

The scarcity of expert-annotated tumor segmentation data has driven adoption of: - **Self-supervised pre-training** (contrastive learning, masked autoencoders) to leverage large unlabeled MRI repositories before fine-tuning on annotated subsets. - **Semi-supervised segmentation** via pseudo-labeling and consistency regularization (mean teacher, virtual adversarial training) applied to BraTS-style data. - **Active learning** frameworks to strategically select the most informative cases for expert annotation, reducing labeling burden by up to 70%.

12.6 Hybrid Spatial-Temporal Frameworks (MONAI 3D ResNet + PINNs)

The most cutting-edge recent advancement (2023-2026) is the hybridization of powerful 3D feature extractors with biophysical mathematical constraints. While traditional models treated segmentation and growth prediction as separate pipelines, modern architectures are fusing them: * **MONAI 3D ResNet Backbones:** Researchers are increasingly utilizing Medical Open Network for AI (MONAI) 3D Residual Networks to process multimodal 3D MRI scans. The residual skip-connections allow these networks to extract incredibly deep spatial features without the vanishing gradient problem, producing highly accurate latent representations of the tumor core and surrounding edema. * **Biophysical PINN Constraints:** Instead of relying purely on mean-squared error against sparse longitudinal data, these architectures feed the 3D ResNet embeddings directly into a **Physics-Informed Neural Network (PINN)**. The PINN loss function acts as a "soft constraint," penalizing the model if its growth predictions violate the Fisher-KPP reaction-diffusion equations. * **Clinical Impact:** This specific hybrid architecture (ResNet + PINN) represents the frontier of neuro-oncology AI. It bridges the gap between raw pixel data and biological reality, allowing for the extraction of patient-specific diffusion (D) and proliferation (ρ) parameters directly from standard imaging, paving the way for biologically consistent, personalized growth simulations.

13. Research Gaps and Future Directions

13.1 Identified Research Gaps

Longitudinal Data Scarcity: The most significant bottleneck in growth modeling. Most datasets contain only pre-operative snapshots. Prospective longitudinal MRI databases with standardized acquisition protocols are critically needed.

Lack of Ground Truth for Growth Prediction: Unlike segmentation (where expert annotation is feasible), validating predicted future tumor maps requires follow-up imaging that may not capture the counterfactual (untreated) growth trajectory.

Integration of Biophysical and Clinical Variables: Most DL models treat tumor growth as a purely imaging phenomenon. Integrating clinical factors (steroid use, chemotherapy cycles, radiation dose mapping) into growth models remains underexplored.

Tumor Micro-environment Modeling: Current DL models do not account for tumor-stroma interactions, immune infiltration, or vessel architecture that critically influence growth patterns.

Generalizability Across Tumor Types: Most models focus exclusively on high-grade glioma (GBM). Meningioma, medulloblastoma, and brain metastases have distinct growth dynamics requiring adapted architectures.

Real-Time Clinical Deployment: Preprocessing pipelines (co-registration, skull stripping, bias-field correction) add significant latency. End-to-end models operating on raw DICOM data without preprocessing are needed.

Uncertainty Quantification: Clinical decisions require not just a prediction, but a confidence interval. Most DL models are deterministic; Bayesian DL and conformal prediction remain underutilized.

Explainability for Clinicians: Despite growing XAI research (Grad-CAM, attention maps), clinicians remain unconvinced by heatmap-based explanations. Physics-informed models like PINNs offer more satisfying biological interpretability.

13.2 Promising Future Directions

- **Hybrid PINN-Transformer architectures**: Combining global context from Transformers with biological growth constraints from PINNs — the next frontier for personalized growth prediction.
- **Diffusion-model-based growth simulators**: Generating patient-specific MRI sequences for any future time point with stochastic uncertainty.
- **Federated longitudinal learning**: Privacy-preserving collaborative growth model training across multiple cancer centers.
- **Digital twin tumor models**: Full patient-specific computational replicas combining MRI, genomics, treatment history, and biophysical PDEs for precision oncology.
- **Foundation models for medical imaging**: Large-scale pre-trained models that generalize to any tumor type, imaging protocol, and downstream task with minimal fine-tuning.

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14. References (IEEE Format)

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Final Summary/Disclaimer

This literature review provides a high-level overview of the significant progress and ongoing challenges in brain tumor growth analysis. The integration of advanced computational techniques like ML and DL has significantly improved our ability to analyze and predict tumor behavior, yet substantial gaps remain, particularly in longitudinal

data coverage and clinical deployment readiness.

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